1. Overview of Computing

1.1 Meaning of Computing

Computing is the process of using computers and related technologies to collect, store, process, and manage information. A computing system uses hardware and software to perform different tasks.

Hardware means the physical parts of a computer, such as the processor, memory, keyboard, and storage devices.

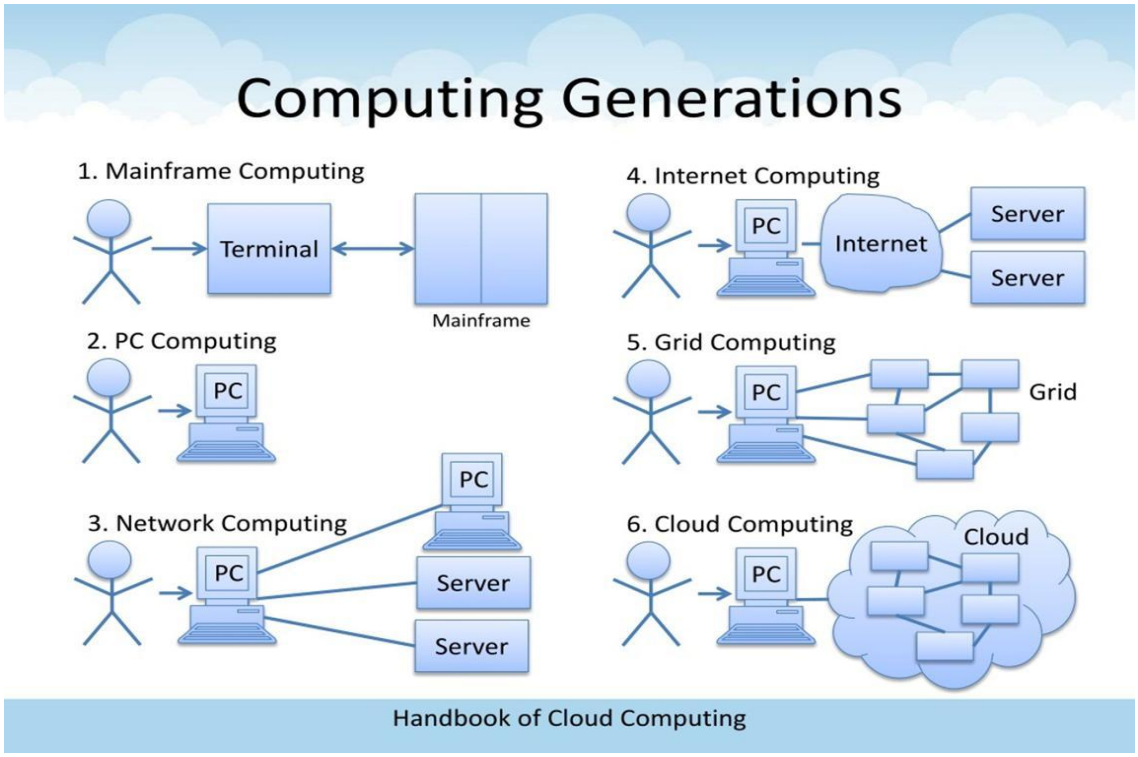
Software means the programs and instructions that tell the computer what to do.

Data means the information that a computer processes, such as numbers, text, images, and videos.

For example, when a student prepares an assignment using a laptop, the laptop receives input from the keyboard, processes the information using software, and displays the result on the screen.

1.2 Development of Computing

Computing has developed through several stages.



1. Personal Computing

Personal computing involves using an individual computer, such as a desktop or laptop, to perform tasks. Examples include writing documents, creating presentations, and programming.

2. Network Computing

Network computing connects two or more computers so that they can exchange information and share resources. For example, computers in a college laboratory may share a printer through a network.

3. Distributed Computing

Distributed computing allows multiple connected computers to work together to complete a task. A large task can be divided into smaller parts and processed by different computers.

4. Virtualised Computing

Virtualisation allows one physical computer to act like multiple separate computers using software. Each virtual computer can run its own operating system and applications.

5. Cloud Computing

Cloud computing allows users to access computing resources, such as storage, servers, and applications, over a network whenever required.

1.3 Important Computing Trends

- Increasing data: Websites, mobile applications, and organisations generate large amounts of data.
- Network connectivity: Computers and devices communicate through networks and the internet.
- On-demand services: Users expect to access computing services whenever they need them.
- Scalability: Computing systems should handle increases or decreases in workload.
- Cost reduction: Organisations try to use resources efficiently and avoid unnecessary hardware expenses.

Example: A student may save files on a laptop, share them over a network, and later store them in cloud storage to access them from different devices.

Key point: Cloud computing builds on earlier technologies such as networking, distributed computing, and virtualisation.

2. Distributed Computing

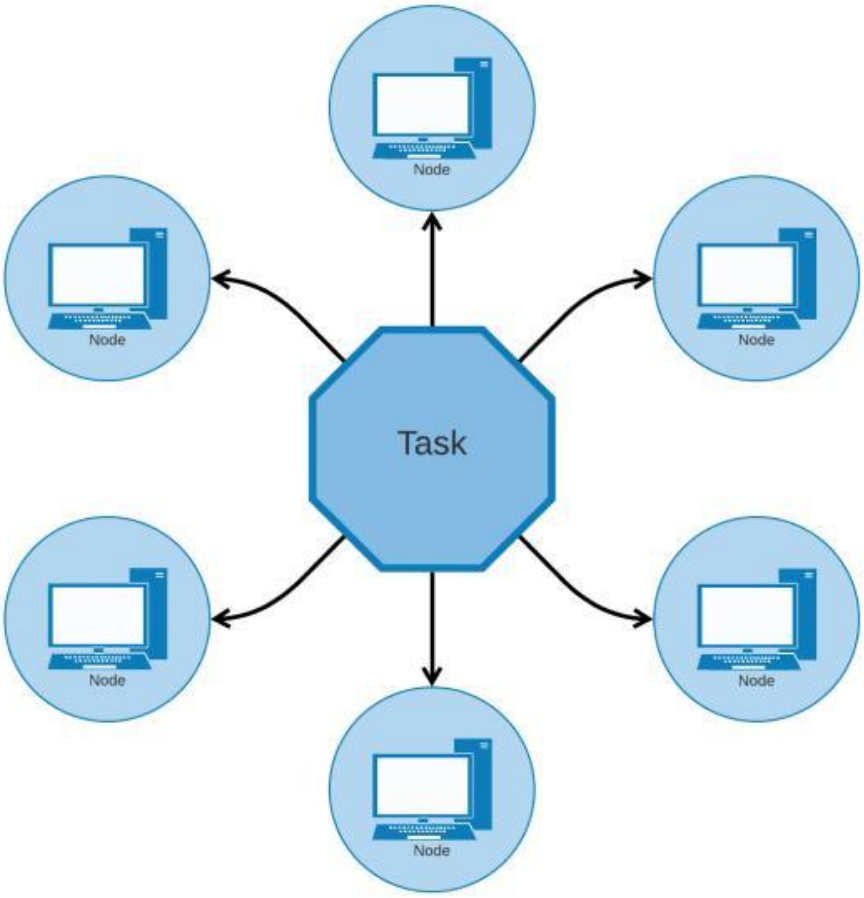
Distributed computing is a computing method in which two or more computers connected through a network work together to complete a task or provide a service.

These computers may be located in the same building or in different geographical locations. They communicate with one another and share work or resources.

2.2 How Distributed Computing Works

Suppose a large task requires a lot of processing power. Instead of assigning the entire task to one computer, the task can be divided into smaller parts.

1. A large task is divided into smaller tasks.
2. The smaller tasks are assigned to different computers.
3. Each computer processes its assigned task.
4. The computers communicate and send their results.
5. The results are combined to produce the final output.



Characteristics of Distributed Computing

1. Multiple computers: The system contains two or more computers.

2. Communication: Computers exchange information through a network.

3. Resource sharing: Computers may share data, storage, software, and processing power.

4. Task division: A large problem can be divided into smaller tasks.

5. Coordination: The computers must coordinate their activities to produce correct results.

2.4 Advantages

- Large tasks can be divided among several computers.
- Resources located in different places can be used together.
- Adding more computers may increase processing capacity.
- The workload can be shared instead of depending on a single computer.

2.5 Disadvantages

- Network failures may interrupt communication.
- Coordinating several computers can be complicated.
- Data consistency and security need careful management.
- Troubleshooting may be more difficult than with a single computer.

2.6 Example

A university wants to analyse thousands of student records. The work can be divided among several computers. Each computer analyses a portion of the records, and the results are combined into one report.

Remember: Distributed computing means that multiple computers cooperate over a network. Cloud computing may use distributed computing, but the two terms are not identical.

3. Grid Computing

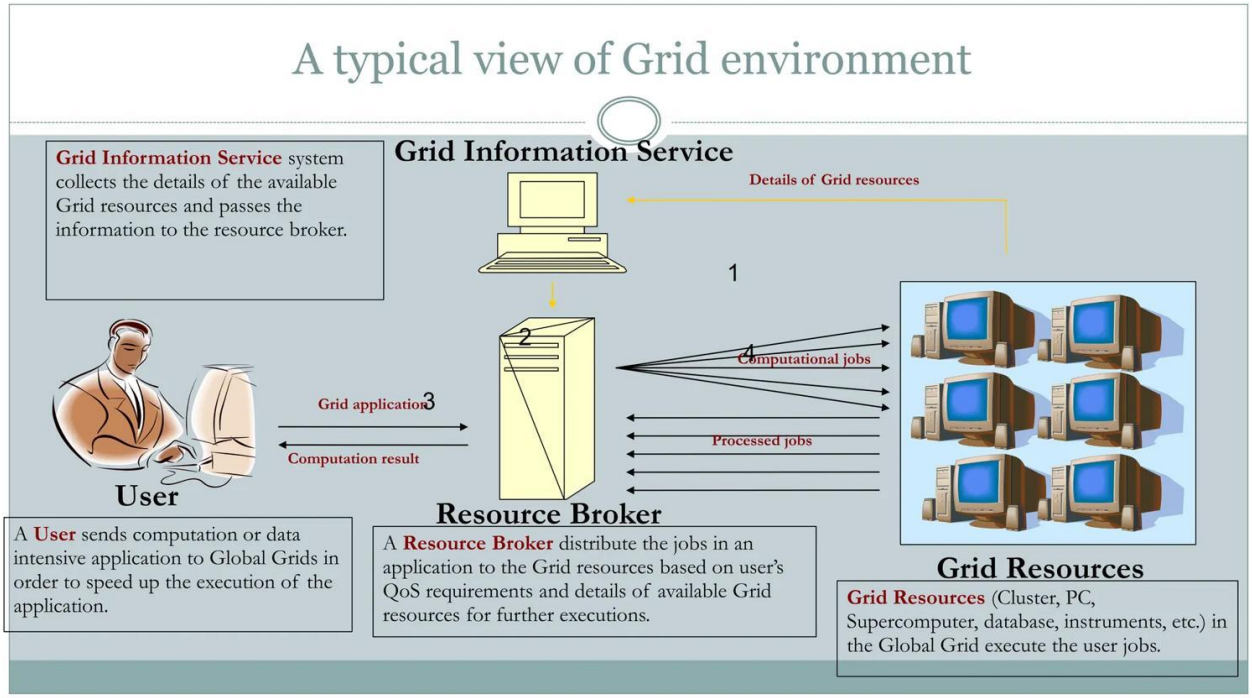
Grid computing is a method of combining and coordinating computing resources from different computers or locations to solve a problem or perform a large task.

The resources may belong to different departments, laboratories, or organisations. Grid software helps coordinate the resources and distribute work.

Grid computing is useful when one computer does not have enough processing power or storage to complete a task efficiently.

3.2 How Grid Computing Works

1. A user submits a large computing task.
2. Grid-management software identifies available resources.
3. The task is divided into smaller jobs when possible.
4. The jobs are assigned to suitable computers.
5. The computers process the jobs.
6. The results are collected and combined



3.3 Types of Grid Computing

1. Computational Grid

A computational grid combines processing resources to perform complex calculations.

Example: Several research centres contribute computing power to scientific calculations.

2. Data Grid

A data grid focuses on sharing, accessing, and managing data stored in different locations.

Example: Research institutions share large scientific datasets.

3. Collaborative Grid

A collaborative grid helps people or organisations share resources and work together on common tasks.

Example: Researchers in different universities collaborate using shared computing resources.

3.4 Advantages

- Shares unused computing resources.
- Helps complete large computing tasks.
- Combines resources from different locations.
- Can reduce the need for one extremely powerful computer.

3.5 Disadvantages

- Requires coordination and management software.
- Network connectivity is important.
- Security and access permissions must be managed carefully.
- Resources owned by different organisations may have different rules.

3.6 Example

Several research laboratories may combine their computing resources to process climate data. Each laboratory processes part of the work, and the results are combined for analysis.

Remember: Grid computing focuses on coordinating and sharing resources that may be spread across different locations or organisations.

4. Cluster Computing

Cluster computing is a method of connecting a group of computers, called nodes, so that they work together as one coordinated system or service.

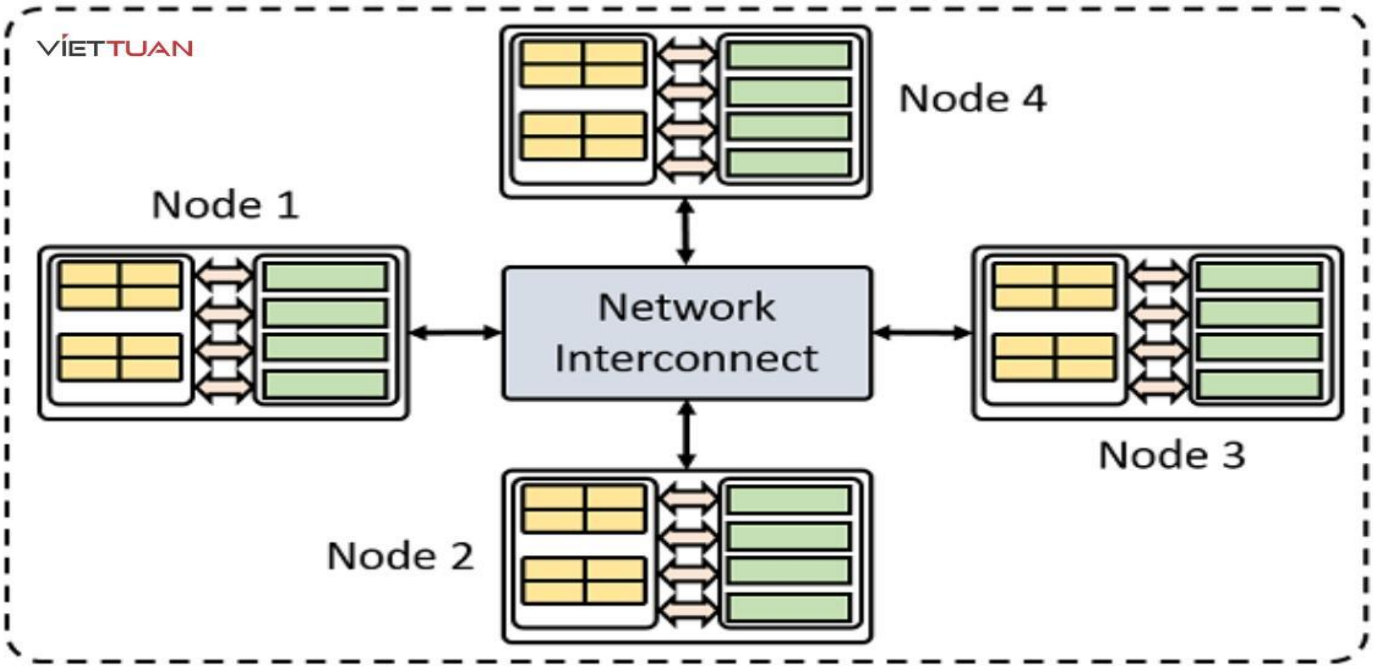
The computers are often located in the same data centre or nearby locations and are connected through a network.

Cluster-management software coordinates the computers and helps them share the workload.

4.2 How Cluster Computing Works

1. Multiple computers are connected through a network.
2. Cluster software coordinates the computers.
3. Tasks or requests are distributed among the nodes.
4. The nodes perform their assigned work.

5. The system combines the results or provides the service



4.3 Types of Cluster Computing

1. High-Performance Computing (HPC) Cluster

An HPC cluster combines the processing power of several computers to solve complex calculations quickly.

Example: Scientific simulations and engineering calculations.

2. High-Availability Cluster

A high-availability cluster is designed to keep a service running even if one computer fails. Another node may take over the service.

Example: A banking service that needs to remain available.

3. Load-Balancing Cluster

A load-balancing cluster distributes incoming requests among multiple computers to prevent one computer from handling all the work.

Example: A website receiving a large number of visitors.

4.4 Advantages

- Provides more processing capacity than a single computer.
- Can improve service availability.
- Distributes workloads among different nodes.
- Can support demanding applications.

4.5 Disadvantages

- Requires proper configuration and monitoring.
- Hardware and network failures may affect performance.
- Maintenance can be complex.
- Initial setup may be costly.

4.6 Difference Between Grid and Cluster Computing

Cluster Computing	**Grid Computing**
Uses a closely coordinated group of computers.	Combines resources that may be spread across locations.
Computers are often managed as one system.	Resources may belong to different organisations.
Often used for high availability and high performance.	Often used to share resources for large computing tasks.
Nodes are usually closely connected.	Resources may communicate over wider networks.

Memory tip: A cluster is a closely coordinated group of computers; a grid combines resources that may be geographically distributed.

5. Utility Computing

Utility computing is a method of providing computing resources as a service, where customers pay according to the resources they use or an agreed service plan.

It is similar to electricity or water services. People use electricity according to their needs and pay the bill based on consumption. In the same way, utility computing allows customers to use computing resources and pay according to the agreed pricing method.

Computing resources may include processing power, storage, software, and network capacity.

5.2 How Utility Computing Works

1. A customer identifies the computing resources needed.
2. A service provider makes those resources available.
3. The customer uses the resources.

4. Usage may be monitored or measured.
5. The customer pays according to the pricing plan.

5.3 Types of Pricing

1. Pay-per-use: The customer pays for the amount of a resource consumed.

2. Subscription-based pricing: The customer pays a fixed fee regularly, such as monthly or yearly.

3. Reserved capacity: The customer pays for resources reserved for a specified period.

4. Tier-based pricing: The price depends on the usage level or service package selected.

5.4 Advantages

- Reduces the need to purchase expensive hardware in advance.
- Makes computing resources available when required.
- Allows customers to pay according to an agreed usage plan.
- Can make it easier to increase computing capacity.
- Reduces some of the burden of managing physical infrastructure.

5.5 Disadvantages

- Costs may increase if resource usage is not monitored.
- Customers may depend on the provider's availability.
- Security and privacy must be considered.
- The customer must understand the provider's pricing and service conditions.

5.6 Relationship Between Utility Computing and Cloud Computing

Utility computing mainly describes providing computing resources as a service, often with usage-based pricing.

Cloud computing is a broader concept. It includes features such as on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.

Cloud computing often uses utility-style pricing, but the two terms do not mean exactly the same thing.

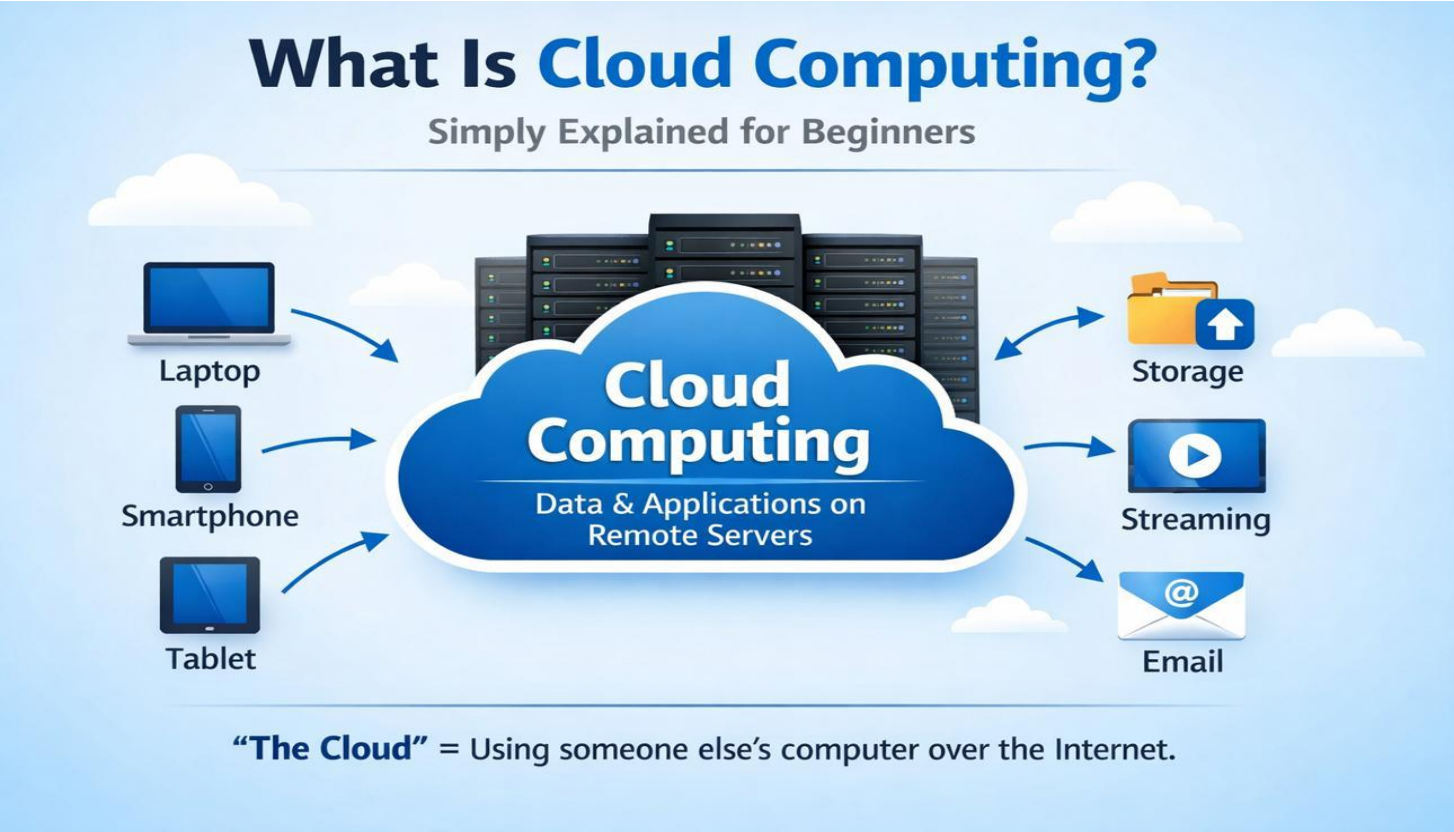
Example: A student or developer rents online storage and pays according to the selected storage plan or amount used.

6. Cloud Computing and the NIST Model

Cloud computing is a model that provides convenient, on-demand network access to a shared pool of configurable computing resources, such as servers, storage, networks, applications, and services. These resources can be provided and released quickly with minimal management effort.

In simple English, cloud computing allows people to use computing resources over a network without having to own and manage all the physical equipment themselves.

For example, instead of buying a physical server for a project, a developer can rent a virtual server from a cloud provider.



6.2 What Is NIST?

NIST stands for the National Institute of Standards and Technology. It provides a widely used definition and framework for cloud computing.

The NIST model identifies five essential characteristics, three service models, and four deployment models.

6.3 Explanation of the NIST Definition

The NIST definition contains several important ideas.

1. On-demand access

Users can request computing resources whenever they need them.

2. Network access

Resources can be accessed through a network, often using the internet.

3. Shared pool of resources

The provider maintains a collection of computing resources that can serve multiple customers.

4. Configurable resources

Users can select suitable resources, such as the amount of storage or computing capacity.

5. Rapid provisioning and release

Resources can be provided and removed quickly as requirements change.

6. Minimal management effort

Users do not need to manage every physical component themselves. The amount they manage depends on the cloud service model.

6.4 Real-Life Examples of Cloud Computing

- Online storage: Saving files using an online storage service.
- Online email: Reading and sending emails through a web browser.
- Cloud servers: Renting virtual servers to host websites.
- Online collaboration: Multiple people editing and sharing documents.
- Cloud-based development: Building, testing, and deploying applications using online platforms.

6.5 Why Is Cloud Computing Important?

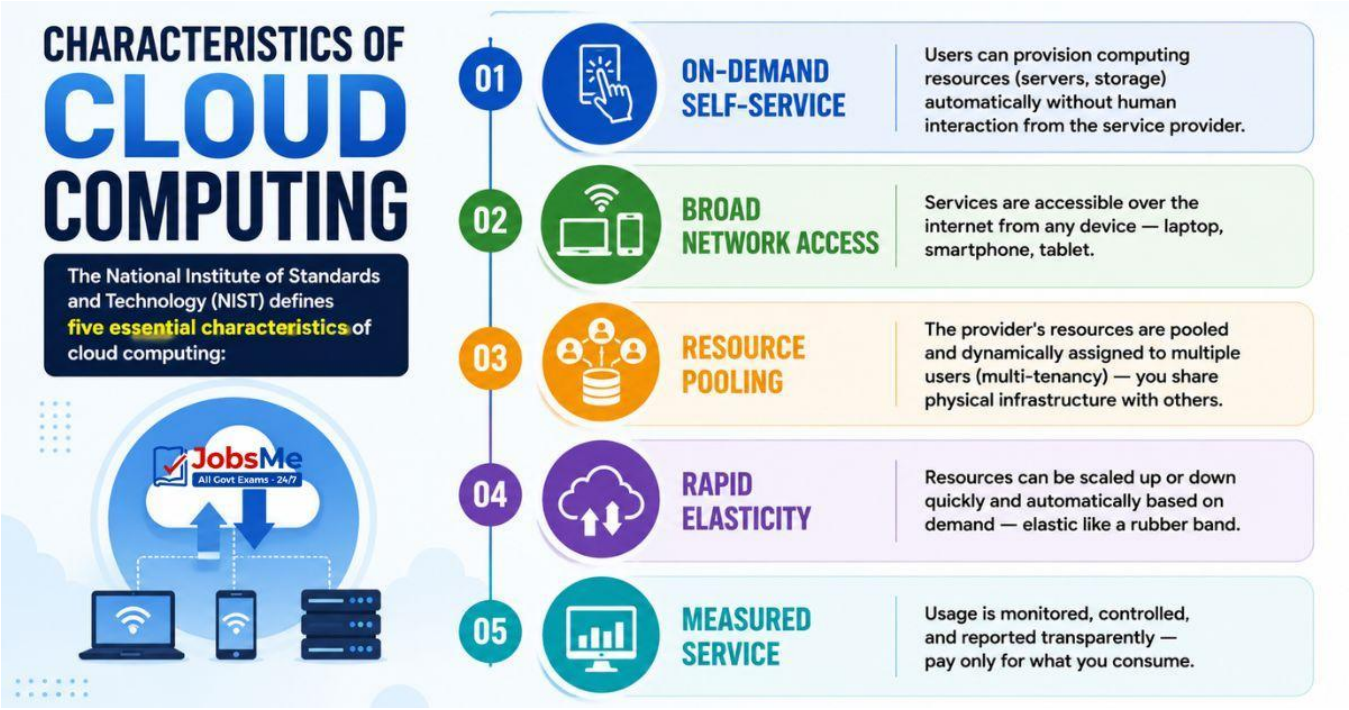
Cloud computing helps individuals and organisations access resources without always purchasing their own infrastructure. It can make it easier to scale applications, share information, deploy services quickly, and manage computing costs.

However, the actual cost and benefits depend on usage, security requirements, provider pricing, and service reliability.

Exam-ready definition: Cloud computing is a model that enables convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort.

7. Essential Characteristics of Cloud Computing

According to the NIST model, cloud computing has five essential characteristics. These are important for understanding how cloud computing works.



7.1 On-Demand Self-Service

On-demand self-service means users can obtain computing resources whenever required without contacting the provider's staff for every individual request.

A user may use an online dashboard or portal to create a virtual server, increase storage, or remove resources.

Example: A developer creates a virtual machine using a cloud provider's website.

Benefit: It saves time and makes resource access more convenient.

7.2 Broad Network Access

Broad network access means cloud services are available through a network and can be accessed using different types of devices, subject to permissions and compatibility.

These devices may include:

- Desktop computers
- Laptops
- Smartphones
- Tablets
- Workstations

Example: A student opens a document on a laptop and later views the same document on a smartphone.

Benefit: Users can access services from different places and devices.

7.3 Resource Pooling

Resource pooling means a cloud provider combines computing resources into a shared pool and allocates them to different customers according to demand.

The provider may share the overall physical infrastructure among multiple customers while maintaining logical separation between their accounts and data.

Customers generally do not choose the exact physical server being used, although they may be able to select a region or other settings.

Example: Many companies use services operated in the same cloud data centre, while their accounts and data remain logically separated.

Benefit: Resources can be used more efficiently.

7.4 Rapid Elasticity

Rapid elasticity means computing resources can be increased or decreased quickly according to demand.

When demand increases, additional resources can be provided. When demand decreases, unnecessary resources can be released.

Example: An online shopping website may need more computing resources during a major sale. After the sale, it can reduce the resources used.

Benefit: It helps applications handle changing workloads and may reduce the cost of unused capacity.

7.5 Measured Service

Measured service means cloud systems monitor and measure resource usage. Measurements may include processing time, storage consumption, or data transfer.

The information can be used for monitoring, billing, and planning.

Example: A cloud dashboard displays the amount of storage consumed and the computing resources used during a month.

Benefit: Customers can monitor usage and understand how resources are being consumed.

7.6 Summary of the Five Characteristics

Characteristic	**Simple meaning**	**Example**
On-demand self-service	Request resources yourself	Create a virtual server
Broad network access	Access services through a network	Open files on a laptop or phone
Resource pooling	Share a managed pool of resources	Multiple customers use a provider's infrastructure
Rapid elasticity	Increase or decrease capacity quickly	Add servers during a busy period
Measured service	Monitor and measure usage	View storage usage and billing

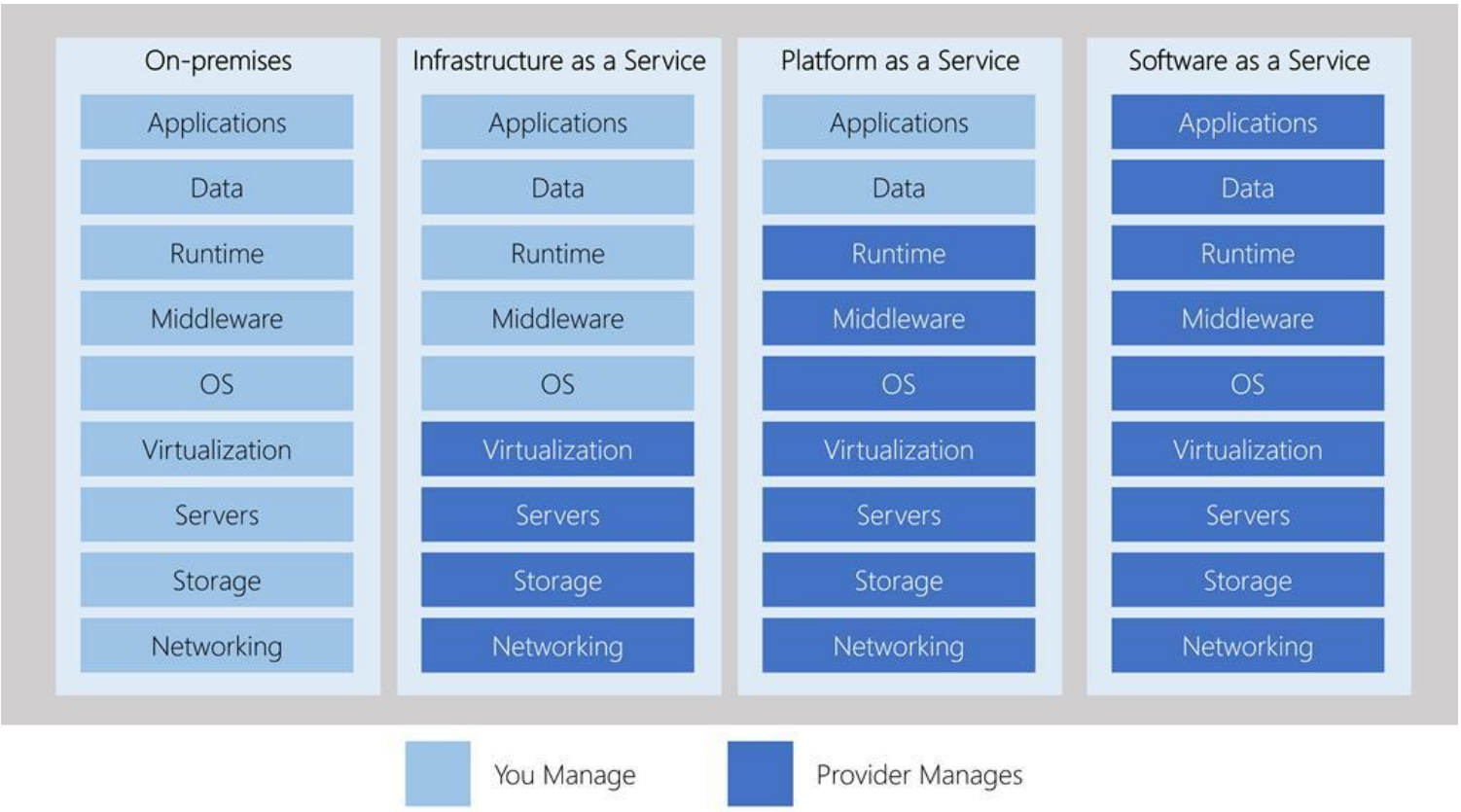
Easy memory tip: Learn these five terms in order: Self-service, Network access, Pooling, Elasticity, Measurement.

8. Cloud Service Models

A cloud service model explains the type of service provided by a cloud provider and the responsibilities of the customer.

The three main service models are:

1. Software as a Service (SaaS)
2. Platform as a Service (PaaS)
3. Infrastructure as a Service (IaaS)



8.1 Software as a Service (SaaS)

Definition

Software as a Service is a cloud model in which users access a complete software application over a network. The provider manages the application and the underlying infrastructure.

Users generally do not need to install or maintain the servers that run the application. They mainly use the software and manage their own content, settings, and access permissions.

Examples

- Online email applications
- Web-based document editors
- Online collaboration tools
- Customer relationship management applications

Advantages

- Applications can often be accessed through a web browser.
- Users do not need to manage the underlying servers.
- The provider handles much of the software maintenance.
- It can make collaboration and sharing easier.

Disadvantages

- Users depend on the provider's service availability.
- Internet access may be required.
- Customisation may be limited compared with managing an application directly.
- Data privacy and access permissions must be considered.

Example: A student uses an online document editor to prepare an assignment instead of installing and managing the entire application infrastructure.

8.2 Platform as a Service (PaaS)

Definition

Platform as a Service provides an environment that developers can use to build, test, deploy, and manage applications.

The provider manages much of the underlying infrastructure and platform. This allows developers to focus more on writing and improving application code.

Examples

- Managed application-hosting platforms
- Cloud development environments
- Platforms that simplify application deployment

Advantages

- Reduces the need to manage physical servers.
- Helps developers build and deploy applications more quickly.
- Provides tools and environments for application development.
- Can support collaboration between developers.

Disadvantages

- Developers may have less control over the underlying environment.
- Some platforms support only particular tools or technologies.
- Moving an application to another platform may require changes.

Example: A student develops a web application and deploys it on a managed cloud platform without configuring every server component manually.

8.3 Infrastructure as a Service (IaaS)

Definition

Infrastructure as a Service provides fundamental computing resources, such as virtual machines, storage, and networking, over a network.

The provider manages the physical infrastructure. The customer has greater control over the operating system, installed software, applications, and configuration of the rented resources.

Examples

- Renting virtual machines
- Cloud storage volumes
- Virtual networks
- Cloud-based computing servers

Advantages

- Provides flexibility and greater control over the software environment.
- Reduces the need to purchase physical servers.
- Resources can often be increased or decreased.
- Supports different operating systems and applications, depending on the service.

Disadvantages

- Customers may need technical knowledge to manage operating systems and applications.
- Configuration mistakes can create security risks.
- Costs need to be monitored carefully.
- The customer remains responsible for many software-level tasks.

Example: A developer rents a virtual server, installs the required software, and uses it to host a website.

8.4 Difference Between SaaS, PaaS, and IaaS

Feature	SaaS	PaaS	IaaS
Full form	Software as a Service	Platform as a Service	Infrastructure as a Service
Service provided	Ready-to-use application	Development and deployment platform	Virtual computing infrastructure
Main user	Application user	Application developer	IT administrator or developer
Customer focus	Using the application	Building and deploying applications	Configuring and managing virtual resources
Management responsibility	Lowest of the three for infrastructure	Shared between provider and developer	Greater customer responsibility
Example	Online email	Managed application platform	Virtual server

Memory tip:

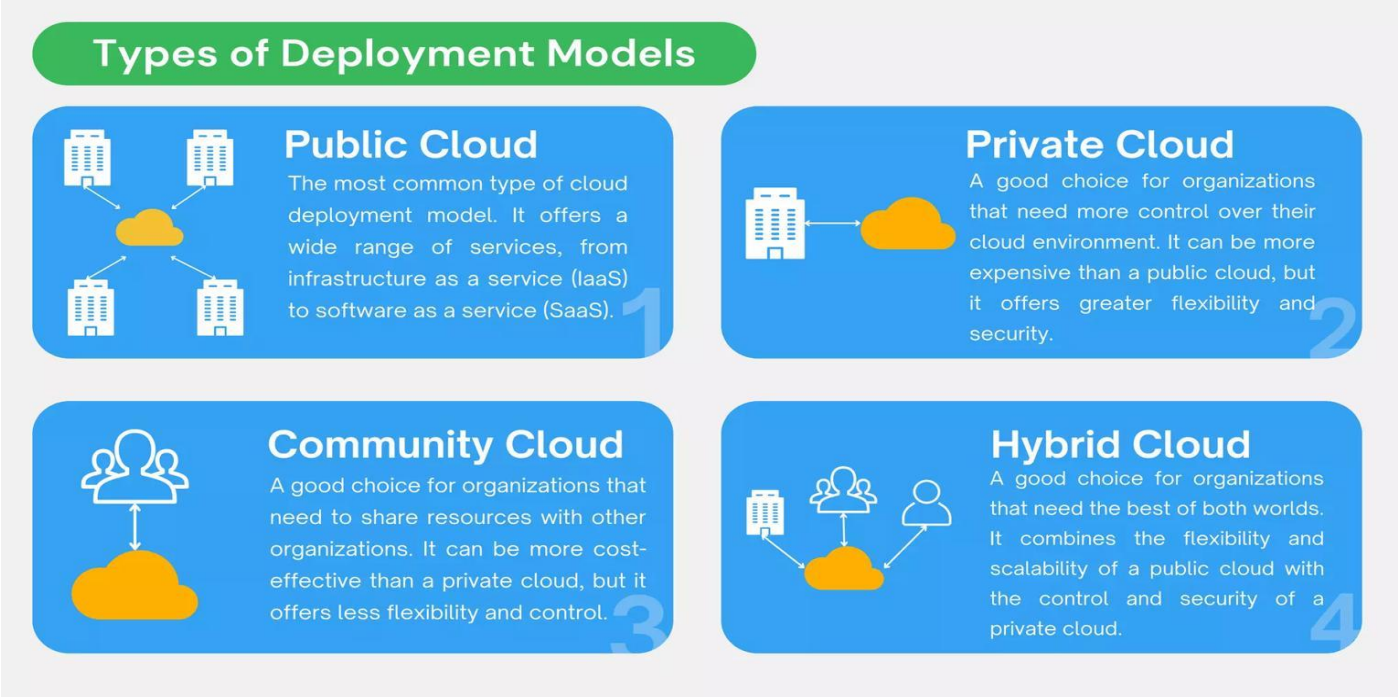
- SaaS: Use software.
- PaaS: Build software.
- IaaS: Rent infrastructure.

9. Cloud Deployment Models

A cloud deployment model describes how a cloud environment is organised, who can use it, and how it is made available.

The four common deployment models in the NIST framework are:

1. Public cloud
2. Private cloud
3. Community cloud
4. Hybrid cloud



9.2 Public Cloud

A public cloud is a cloud environment made available to the public or a large group of customers through a service provider.

The provider operates the infrastructure and offers services to multiple customers. Customers use the services according to their accounts and permissions.

Advantages

- Usually avoids the need to purchase physical infrastructure.
- Resources can often be obtained quickly.
- Capacity can be increased according to demand.
- The provider manages the underlying physical infrastructure.

Disadvantages

- Customers have less direct control over the physical infrastructure.
- Security and privacy require proper configuration and management.
- Costs can increase if usage is not monitored.
- Customers depend on the provider's service availability.

Example: A company rents virtual servers from a commercial cloud provider to host its website.

9.3 Private Cloud

A private cloud is used exclusively by one organisation. It may be managed by the organisation itself or by a third party. It may be located on the organisation's premises or elsewhere.

Private clouds can offer more control over configurations, access policies, and resource management.

Advantages

- Supports organisation-specific security policies.
- Provides greater control over the cloud environment.
- Can be configured for particular organisational requirements.

Disadvantages

- Setting up and maintaining the environment may be expensive.
- Technical staff may be needed for management.
- The organisation must plan for capacity, security, and maintenance.

Example: A large institution operates a cloud environment exclusively for its own departments.

9.4 Community Cloud

A community cloud is shared by several organisations that have common requirements, such as security rules, policies, or business objectives.

It may be managed by one or more of the participating organisations or by a third-party provider.

Advantages

- Allows organisations with similar needs to share resources.
- Can support common security and operational requirements.
- May reduce costs compared with each organisation maintaining a separate environment.

Disadvantages

- Organisations must agree on management and access policies.
- Costs and responsibilities must be shared fairly.
- Coordination between the participating organisations can be difficult.

Example: Several organisations with similar regulatory requirements share a cloud environment.

9.5 Hybrid Cloud

A hybrid cloud combines two or more distinct cloud infrastructures, such as private and public clouds. They remain separate environments but are connected so that data or applications can move between them when appropriate.

Advantages

- Combines different cloud environments according to their strengths.
- Allows an organisation to keep some workloads in a more controlled environment.
- Can provide extra capacity when demand increases.

Disadvantages

- Connecting the environments can be technically complex.
- Security policies must be coordinated.
- Managing data movement and costs requires planning.

Example: A business keeps sensitive information in a private environment but uses public-cloud resources when its website receives more visitors than usual.

9.6 Comparison of Deployment Models

Model	Who uses it?	Main idea
Public cloud	Multiple customers	Provider offers cloud services to many customers
Private cloud	One organisation	Cloud environment used exclusively by one organisation
Community cloud	Organisations with shared requirements	Environment shared by a defined community
Hybrid cloud	An organisation using combined environments	Connects separate cloud infrastructures

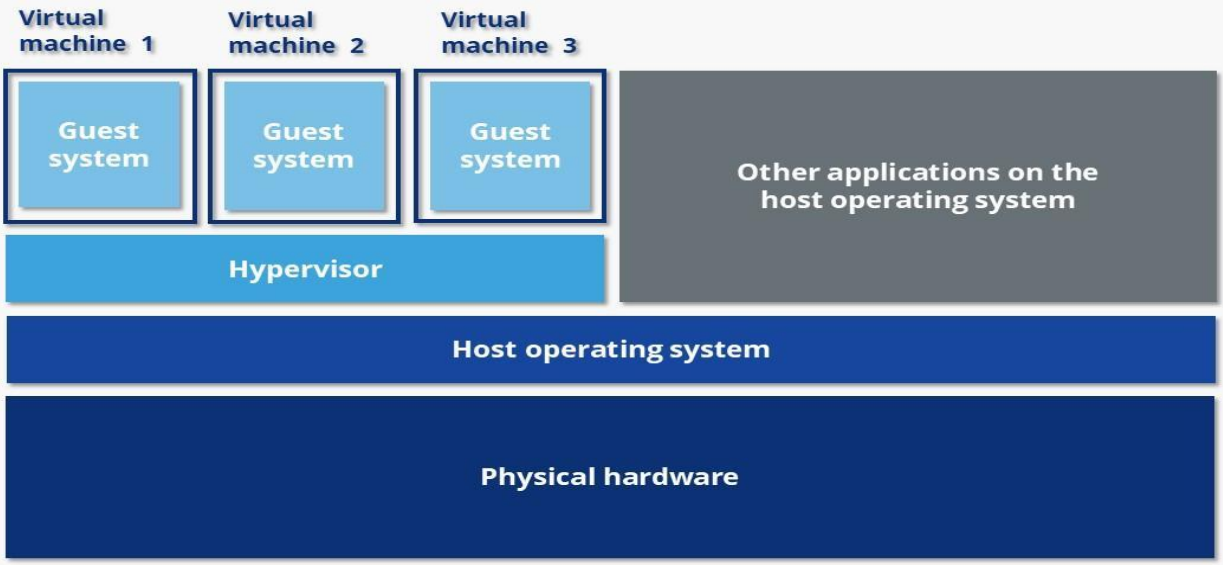
Memory tip: Public means many customers, private means one organisation, community means shared needs, and hybrid means a combination.

10. Cloud Computing and Virtualisation

Virtualisation is a technology that creates software-based versions of computing resources. It allows one physical computer to provide multiple virtual computing environments.

For example, one physical server can run several virtual machines. Each virtual machine can have its own operating system and applications.

Virtualisation is an important technology used by many cloud providers to allocate and manage computing resources.



10.2 Important Terms

1. Physical Host

The physical host is the actual hardware computer or server. It contains components such as the processor, memory, storage, and network interfaces.

2. Virtual Machine (VM)

A virtual machine is a software-based computer that receives virtual computing resources. It can run an operating system and applications like a physical computer.

3. Hypervisor

A hypervisor is software or a system layer that creates and manages virtual machines. It allocates physical resources to the virtual machines.

4. Guest Operating System

The guest operating system is the operating system installed inside a virtual machine. Different VMs may run different operating systems, depending on the hypervisor and its support.

10.3 How Virtualisation Works

1. A physical server provides computing resources.
2. A hypervisor manages the available resources.
3. The hypervisor creates virtual machines.
4. Each virtual machine receives allocated CPU, memory, storage, and network resources.
5. Each virtual machine runs its operating system and applications.

This arrangement allows several virtual machines to use the resources of one physical server.

10.4 Advantages of Virtualisation

- Better resource utilisation: Multiple virtual machines can use the same physical server.
- Flexibility: Virtual machines can be created or removed using software.
- Isolation: Workloads can be separated into different virtual environments.
- Cost savings: Organisations may need fewer physical servers for suitable workloads.
- Simplified management: Virtual resources can often be managed through software tools.

10.5 Disadvantages of Virtualisation

- If the physical host fails, several virtual machines running on it may be affected.
- Poor resource allocation can reduce performance.
- Virtualisation requires appropriate software and technical management.
- Incorrect configuration can create security risks.

10.6 Difference Between Virtualisation and Cloud Computing

Virtualisation	**Cloud Computing**
A technology for creating virtual computing resources.	A model for delivering computing resources as services.
Can allow several VMs to run on one physical server.	Provides on-demand access to a shared pool of resources.
Does not automatically provide self-service or measured billing.	Includes characteristics such as self-service, elasticity, and measured service.
Can be used without offering cloud services.	May use virtualisation as one of its supporting technologies.

Important point: Virtualisation helps enable cloud computing, but virtualisation alone does not make a system a cloud.

11. Advantages and Disadvantages of Cloud Computing

Cloud computing provides flexible access to computing resources. However, it also introduces risks and responsibilities that users and organisations must consider.

11.1 Advantages of Cloud Computing

1. Lower Initial Investment

Organisations may avoid purchasing all the physical servers, storage systems, and networking equipment in advance. They can obtain resources from a provider instead.

2. Scalability

Cloud resources can be increased when an organisation needs more capacity.

For example, a website can obtain additional computing resources as its number of visitors increases.

3. Accessibility

Cloud services can often be accessed from different places and devices through a network, provided users have the required permissions and connectivity.

4. Rapid Provisioning

Computing resources can often be created much faster than purchasing and installing new physical hardware.

5. Reduced Physical Infrastructure Maintenance

The provider handles many physical infrastructure tasks. The exact responsibilities of the customer depend on the service model.

6. Measured Usage

Usage information can help organisations understand how much computing capacity they consume and what they may need to pay.

7. Collaboration

Cloud-based applications can help people share files, communicate, and work on common documents.

8. Flexible Resource Management

Resources can often be increased, reduced, or released according to changing requirements.

11.2 Disadvantages of Cloud Computing

1. Internet and Network Dependence

Cloud services depend on network connectivity. Poor connectivity can make services slow or unavailable.

2**. Security and Privacy Risks**

Data stored or processed in the cloud must be protected. Weak passwords, incorrect permissions, and poor configuration can create security problems.

3. Service Outages

If a provider experiences a failure, customers may temporarily lose access to a service.

4. Unpredictable Costs

Costs may increase if users leave unnecessary resources running or do not monitor storage, computing, and data-transfer charges.

5. Vendor Lock-In

Vendor lock-in occurs when moving from one provider to another becomes difficult or expensive because of technical differences, proprietary features, or data-transfer requirements.

6. Less Direct Control

Customers generally do not directly manage the provider's physical servers and networking equipment.

7. Legal and Compliance Requirements

Some organisations have rules about where data may be stored, who can access it, and how it must be protected.

11.3 Advantages vs Disadvantages

Advantages	Disadvantages
Lower initial investment	Costs can increase with usage
Flexible capacity	Network dependence
Quick resource provisioning	Provider outages
Access from different locations	Security and privacy concerns
Less physical infrastructure maintenance	Less direct hardware control
Usage monitoring	Vendor lock-in may make migration difficult

11.4 How to Reduce Cloud Risks

Organisations can reduce risks by:

- Using strong authentication and access controls.
- Encrypting sensitive information where appropriate.
- Monitoring resource usage and costs.
- Maintaining suitable backups and recovery plans.
- Reviewing the provider's service terms and SLA.
- Planning how to move data or applications if the provider changes.

Conclusion: Cloud computing can improve flexibility and resource efficiency, but it should be adopted after considering costs, security, reliability, and organisational needs.

12. Cloud Adoption and Economics

12.1 Meaning of Cloud Adoption

Cloud adoption means moving some or all computing activities from traditional systems to cloud-based services.

An organisation may use cloud computing for storing files, hosting websites, developing applications, analysing data, or running business software.

Cloud adoption does not always mean moving everything to the cloud. Some organisations use a combination of cloud services and their existing systems.

12.2 Reasons for Cloud Adoption

1. Cost Management: Organisations may avoid buying all the hardware required at the beginning.

2. Scalability: Computing resources can be adjusted as demand changes.

3. Faster Deployment: New servers and services can often be created quickly.

4. Flexibility: Organisations can select services according to their requirements.

5. Collaboration: Employees can access and share information through suitable cloud applications.

12.3 Cloud Economics

Cloud economics is the study of the costs and benefits of using cloud computing compared with other computing arrangements.

An organisation should consider the total cost of using a service rather than looking only at its advertised price.

Important Cost Factors

- Computing resources, such as virtual machines and processors.
- Storage capacity and storage performance.
- Network traffic and data transfer.
- Software licences and support plans.
- Backup, monitoring, and security services.
- Migration, training, and staff time.
- Costs associated with moving to another provider.

12.4 How Elasticity Can Help Control Costs

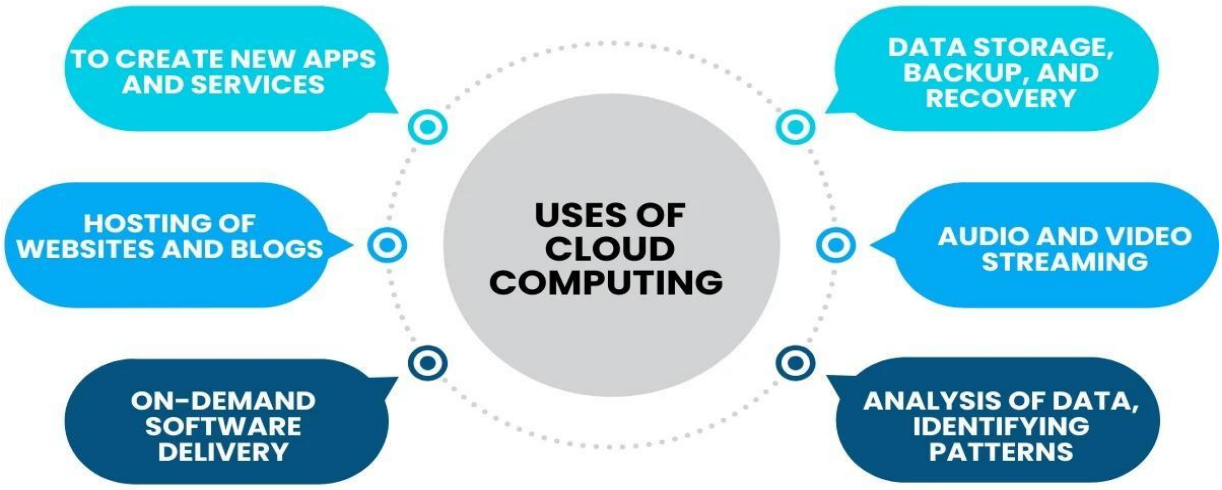
Suppose an online shopping website normally receives 1,000 visitors per day but receives 20,000 visitors during a festival sale.

The website may need additional computing resources during the sale. After the sale, it can reduce those resources.

This is useful because the organisation does not necessarily need to maintain peak capacity all the time.

However, cloud computing does not automatically guarantee lower costs. Organisations must monitor usage and remove or resize unnecessary resources.

12.5 Common Uses of Cloud Computing



Cloud computing is used for:

1. Website hosting: Running websites and web applications.
2. Online storage: Storing and sharing documents, photos, and other files.
3. Software development: Building, testing, and deploying applications.
4. Data analytics: Processing large datasets to find useful information.
5. Artificial Intelligence and Machine Learning: Obtaining computing resources for training or running models.
6. Online collaboration: Sharing documents and working with team members.
7. Backup and recovery: Maintaining copies of data and supporting recovery after failures.
8. Online education: Providing learning platforms, recorded lectures, and educational resources.

13. Service Level Agreement (SLA)

A Service Level Agreement, commonly called an SLA, is an agreement that describes the expected level of service and the responsibilities of the service provider and customer.

In cloud computing, an SLA may describe service availability, support arrangements, response times, and what happens if agreed service levels are not met.

13.2 Important Elements of an SLA

1. Service Availability

It describes how available a service is expected to be during an agreed period.

2. Performance

It may describe expected service performance or response times.

3. Customer Support

It explains how customers can obtain help and what support is included.

4. Responsibilities

It identifies the responsibilities of the provider and the customer.

5. Remedies

It may describe the remedies available if the provider does not meet particular commitments.

13.3 Importance of an SLA

- It clarifies the expected service level.
- It helps customers understand the provider's responsibilities.
- It provides a reference for evaluating service performance.
- It explains what support and remedies may be available.

Example: A company checks a cloud provider's SLA before hosting an important website. It reviews the availability commitment and the support arrangements offered by the provider.

Important: An SLA does not eliminate the need for backups, security measures, and disaster-recovery planning.

14. Role of Open Standards in Cloud Computing

Source note: Your Week-01 lecture outline includes the role of open standards, but the supplied slides give limited detail on this topic. The following is a basic explanation to help you understand the concept.

14.1 Definition

Open standards are publicly available technical specifications that help different systems, applications, and organisations communicate and work together.

They may define common data formats, communication protocols, interfaces, or methods for exchanging information.

In cloud computing, open standards can help different software systems and services interact more easily.

14.2 Importance of Open Standards

1. Interoperability

Interoperability means that different systems can communicate and exchange information.

For example, a cloud application may exchange information with another application through a compatible interface.

2. Portability

Portability means that data or applications can be moved from one environment to another.

Standard formats and interfaces may make this process easier, although some migration work may still be required.

3. Reduced Vendor Dependence

Open standards can reduce dependence on a single provider's proprietary methods.

However, they do not completely prevent vendor lock-in.

4. Easier Integration

Open standards can help organisations connect cloud services with their existing applications and systems.

14.3 Example

Suppose a company uses one software system to manage customer information and another system to prepare reports.

If both systems support compatible data formats and interfaces, exchanging information between them may become easier.

14.4 Limitations

- Not every cloud provider supports the same features.
- Proprietary technologies may create compatibility problems.
- Moving applications between providers may require changes.

- Security and access policies must still be managed correctly.

Conclusion: Open standards support interoperability, portability, and integration. They can make cloud environments easier to connect, but they do not guarantee that every service will work identically.

15. Important Differences for Revision

These comparisons are useful for short-answer and long-answer examinations.

15.1 Distributed, Grid, and Cluster Computing

Point	Distributed	Grid	Cluster
Meaning	Multiple computers cooperate over a network	Resources from different locations are coordinated	Closely coordinated computers work as one system or service
Location	May be in different locations	May be spread across locations and organisations	Often located in one site
Main purpose	Share work and resources	Combine distributed resources for large tasks	Improve processing, availability, or load handling
Management	Depends on the system	May involve different administrative authorities	Usually managed as one coordinated system

15.2 Cloud Service Models

SaaS	PaaS	IaaS
Software as a Service	Platform as a Service	Infrastructure as a Service
Provides a ready-to-use application	Provides a platform for building and deploying applications	Provides virtual computing infrastructure
User mainly uses the application	Developer mainly works on application code	Customer manages more of the software environment
Example: Online email	Example: Managed application platform	Example: Rented virtual server

15.3 Cloud Deployment Models

Public Cloud	Private Cloud	Community Cloud	Hybrid Cloud
Available to multiple customers	Used exclusively by one organisation	Shared by organisations with common needs	Combines distinct cloud environments
Provider offers shared services	Dedicated to one organisation	Shared management and requirements	Environments are connected for a purpose

16. Important Questions with Answers

2-Mark Questions

1. What is cloud computing?

Cloud computing provides on-demand network access to a shared pool of configurable computing resources that can be rapidly provided and released with minimal management effort.

2. What is distributed computing?

Distributed computing is a method in which multiple networked computers work together to complete a task or provide a service.

3. What is grid computing?

Grid computing combines computing resources from different locations or organisations to perform large computing tasks.

4. What is cluster computing?

Cluster computing connects a group of computers called nodes so that they work together as one coordinated system or service.

5. What is utility computing?

Utility computing provides computing resources as a service, often with charges based on usage or an agreed service plan.

6. What are the five essential characteristics of cloud computing?

On-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.

7. What is virtualisation?

Virtualisation is a technology that creates software-based versions of computing resources, such as virtual machines.

8. What is a hypervisor?

A hypervisor is software or a system layer that creates, manages, and allocates resources to virtual machines.

9. What is SaaS?

Software as a Service provides a complete application that users can access over a network.

10. What is PaaS?

Platform as a Service provides an environment for developing, testing, and deploying applications.

11. What is IaaS?

Infrastructure as a Service provides virtual computing resources such as servers, storage, and networking.

12. What is an SLA?

A Service Level Agreement defines expected service levels and the responsibilities of the provider and customer.